

Tessa Cannon

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EDUCATION

University College London, London, UK — MSc in **Artificial Intelligence** for Sustainable Development
2024 - 2025

First Class Honors

Rice University, Houston, TX, USA — BA in **Political Science**, Minors: **Statistics & Business**

IES Study Abroad, Paris, FR — Business & International Affairs

2018 - 2022

3.933/4.0 GPA, *Magna Cum Laude*

Lincoln High School, Portland, OR, USA — International Baccalaureate

2014 - 2018

4.0/4.0 GPA, *Valedictorian*

RELEVANT COURSEWORK

Deep Representations and Learning, Probabilistic Modeling, Reinforcement Learning, Computer Vision, Foundations of AI, Linear Algebra, Linear Regression, AI for Sustainable Development, Applied ML and Data Science, AI for Earth Observation

EXPERIENCE

Valtech, Portland, OR — **Data Science Analyst**

July 2022 - June 2024

- Leveraged **SQL** on **AWS Redshift** to generate actionable insights from raw data for a **Fortune Global 500** client.
- Optimized marketing channels by building a **Dockerized ETL** pipeline, delivering an estimated **12x ROI**.
- Fine-tuned **Prophet time series** forecast to model complex seasonal sales patterns across diverse product types.
- Created automated marketing dashboards (**Tableau, Looker, PBI**) to surface insights from siloed data sources.

Google-Rice University Data Science Research Experience, Houston, TX — **Data Science Research Intern**

May 2021 - July 2021

- Built a **PyTorch-based DenseNet-121** model for remote sensing classification of housing types using density maps.
- Engineered custom datasets by preprocessing 13,000 high-resolution aerial image tiles with ArcGIS and Python.
- Earned **1st Runner-Up prize** for a research presentation judged by Google engineers and Rice University faculty.

ADHD Child & Family Services, Portland, OR — **Office Administrator**

Rice University, Houston, TX — **Political Science PhD Research Assistant**

Oregon League of Conservation Voters, Portland, OR — **Intern**

PROJECTS

Dissertation on Generating 3D Topography Using Satellite-Conditioned Diffusion — University College London

- Built **30M-parameter** conditional diffusion model to generate synthetic topography from satellite images of Arctic ice.
- Engineered a modular **UNet-based** diffusion framework in PyTorch with **self-attention** layers and **permutation-invariant** conditioning, enabling generalization across unseen Arctic regions.
- Automated geolocation of **Sentinel-2** and **LiDAR GeoTIFFs** and built PyTorch dataloaders for patch-based training.
- Designed a suite of physical and statistical evaluation metrics (e.g., **JSD, PSD**) to quantify realism of generated maps.
- Tracked and analyzed model performance with **Weights & Biases**, running **parallel ablation studies** across multiple RTX 4070 Ti GPUs to evaluate performance of architecture and noise-schedule variants.

Variational Autoencoder for Robotic Behavior Cloning — University College London

- Trained a **VAE** for robotic trajectory prediction; achieved **80% prediction accuracy** on latent-encoded image inputs.
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ACTIVITIES

Data Science Senior Capstone Project, Houston, TX — **Senior Team Member**

January 2022 - May 2022

- Developed a predictive fire risk model and interactive property risk dashboard, enabling the Houston Fire Department to streamline fire inspection and prevention in order to improve efficiency of property inspections and save lives.

Rice University Students Demand Action, Houston, TX — **President and Founder**

December 2019 - May 2021

- Founded Rice University Chapter of Students Demand Action to reduce gun violence in the state of Texas.
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SKILLS

Languages — native English, fluent French

Tools and Software — Python, SQL, R, PyTorch, W&B, AWS, Docker, Tableau, Google Big Query, Looker Studio, ArcGIS

AWARDS

- Rice University President's Honor Roll
- 1st Runner Up Poster Presentation at Google Data Science Research Internship
- National Merit Scholarship Winner